

Importing required libraries

```
import urllib.request # Handling URL
```

- Helps you open a website link and read its content.

```
from bs4 import BeautifulSoup # Handling or parsing html files
```

- BeautifulSoup is used to extract text from HTML pages.

```
import nltk # toolkit
```

- NLTK = Natural Language Toolkit → used for text processing.

```
nltk.download('stopwords') # or is was extracted
```

- Downloads common useless words like “the”, “is”, “and”.

```
from nltk.corpus import stopwords
```

- Imports the stopwords list into your program.

2. Getting information from a website

```
url = "https://en.wikipedia.org/wiki/Tata_Group"
```

- Opens the Wikipedia page for TATA.

```
req = Request(url, headers={'User-Agent': 'Mozilla/5.0'})
```

- Wikipedia is rejecting the request because no User-Agent header is provided. Because the website thinks you are a normal browser. Wikipedia protects its server. When Python requests look like this:
User-Agent: Python-urllib/3.9. They block it as a bot → 403 Forbidden.

By adding:

```
'User-Agent': 'Mozilla/5.0'
```

You look like Chrome browser → allowed.

```
html = response.read()
```

- Reads the full HTML content of the webpage.

```
soup = BeautifulSoup(html, 'html5lib')
```

- Converts HTML into a structured format so we can extract text easily.

```
text = soup.get_text(strip=True)
```

- Pulls all text from the webpage and removes extra spaces.

4. Tokenization (splitting into words)

```
tokens = [t for t in text.split()]
```

- Splits the text into individual words (tokens).

Example:

"Hello world" → ["Hello", "world"]

5. Removing stopwords

```
sr = stopwords.words('english')
```

- Loads English stopwords (like “the”, “and”, “is”).

```
clean_tokens = tokens[:]
```

- Makes a copy of tokens to clean.

```
for token in tokens:  
    if token in stopwords.words('english'):  
        clean_tokens.remove(token)
```

- Loops through each word.
- If the word is a stopword → remove it from clean_tokens.

Result:

Only meaningful words remain → no “the”, “is”, “and”, etc.

6. Frequency distribution of words

```
freq = nltk.FreqDist(clean_tokens)
```

- Counts how many times each word appears.

Example:

```
{"TATA": 20, "company": 10, "India": 5}
```

7. Plotting the most common words

```
for key, val in freq.items():  
    freq.plot(50)
```

- Plots the top 50 most frequent words in a bar graph.
- Only the first iteration matters because the plot is called inside the loop → but graph shows only once.

```
## print(str(key) + ':' + str(val))
```

- (Commented) Would print each word and its count.